

ENGINEERING MECHANICS

UNIT II: TWO-DIMENSIONAL FORCE SYSTEMS & FRICTION

Harcourt Butler Technical University (HBTU), Kanpur — B.Tech 1st Year
Complete Theoretical Concepts, Analytical Formulations, Laws, Free-Body Diagrams, and Solved Frameworks

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1 Part 1: Two-Dimensional Force Systems

1.1 Basic Concepts

Engineering Mechanics deals with the state of rest or motion of bodies subjected to the action of forces. In statics, we examine rigid bodies under equilibrium conditions.

- **Rigid Body:** An idealization of a solid body in which deformation is negligible under the applied load; the relative distance between any two constituent particles remains constant under all external forces.
- **Force:** A vector quantity representing the interaction between bodies that tends to change or alters the state of rest or uniform motion. It is fully defined by four characteristics:
 1. *Magnitude* (e.g., in Newtons, N or kN).
 2. *Direction* (line of action and angle θ with respect to a reference axis).
 3. *Sense* (indicated by the arrowhead along the line of action).
 4. *Point of Application* (location on the body where the force is exerted).
- **Classification of 2D Coplanar Force Systems:**
 - *Concurrent:* Lines of action of all forces pass through a single common point.
 - *Non-concurrent:* Lines of action do not intersect at a single point.
 - *Parallel:* Lines of action of all forces are parallel to each other (like or unlike parallel).
 - *Collinear:* Forces act along the exact same line of action.

1.2 Newton's Laws of Motion

Newtonian mechanics forms the foundational bedrock of classical engineering mechanics:

1. **Newton's First Law (Law of Inertia):** Every body continues in its state of rest, or of uniform motion in a straight line, unless acted upon by a net external unbalanced force. In statics: $\sum \vec{F} = \vec{0} \implies \vec{a} = \vec{0}$.
2. **Newton's Second Law (Law of Acceleration):** The rate of change of momentum of a body is proportional to the net applied force and takes place in the direction of the force:

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = m\vec{a} \quad (\text{for constant mass } m) \quad (1)$$

3. **Newton's Third Law (Action & Reaction):** To every action, there is an equal and opposite reaction. If body A exerts a force $\vec{F}_{AB}$ on body B , then body B exerts an equal, collinear, and opposite force $\vec{F}_{BA} = -\vec{F}_{AB}$ on body A .

1.3 Principle of Transmissibility of Forces

Principle Statement

The state of rest or external motion of a rigid body remains unaltered if a force $\vec{F}$ acting at a point A is replaced by an equal and identical force $\vec{F}$ acting at any other point B along the same line of action.

- **Mathematical Representation:** Force is treated as a *sliding vector*.
- **Key Limitation:** Transmissibility applies strictly to **external rigid-body effects** (reactions, accelerations, overall equilibrium). It **does not** apply to **internal stresses, strains, or deformations**.
- **Example:** Pushing a trolley from behind with force F produces the identical external acceleration as pulling it from the front with force F along the same line. However, the push causes internal compression in the frame, whereas the pull causes tension.

1.4 1.4 Transfer of a Force to a Parallel Position (Force-Couple System)

In engineering analysis, it is frequently necessary to shift the line of action of a force $\vec{F}$ from a point A to a parallel position passing through another point B separated by a perpendicular distance d .

- Applying two equal and opposite forces $+\vec{F}$ and $-\vec{F}$ at point B does not alter the external effect on the body since the net added force is zero.
- The original force $\vec{F}$ at A and the opposite force $-\vec{F}$ at B form a **couple** of magnitude:

$$M = F \cdot d \quad (2)$$

where the direction (clockwise or counter-clockwise) is determined by the sense of rotation about B .

- The remaining force $+\vec{F}$ at B maintains the translational tendency.
- **Conclusion:** A force $\vec{F}$ acting at point A can be replaced at an arbitrary point B by an identical parallel force $\vec{F}$ acting at B together with a couple of moment $M = F \cdot d$.

1.5 1.5 Moment of a Force and Varignon's Theorem

- **Moment of a Force:** The turning or rotational effect of a force about an axis or point O is defined as:

$$\vec{M}_O = \vec{r} \times \vec{F} \implies M_O = F \cdot d \quad (3)$$

where $\vec{r}$ is the position vector from O to any point on the line of action of $\vec{F}$, and $d = r \sin \theta$ is the perpendicular moment arm.

- **Varignon's Theorem (Principle of Moments):** The algebraic sum of the moments of a system of coplanar forces about any point is equal to the moment of their resultant force about that same point:

$$M_O = \sum_{i=1}^n M_{O,i} = \sum (\vec{r}_i \times \vec{F}_i) = \vec{r} \times \vec{R} \quad (4)$$

In scalar Cartesian coordinates: $M_O = \sum (x_i Y_i - y_i X_i) = x R_y - y R_x$.

1.6 1.6 Resultant of Coplanar Force Systems

The resultant $\vec{R}$ is a single equivalent force (or force-couple) that produces the exact same external dynamic and static effect as the entire system of forces.

1.6.1 A. Concurrent Force System

Since all lines of action intersect at a common origin/point O , the resultant passes directly through O (no residual couple).

Formulation for Concurrent System

Resolve all forces into orthogonal Cartesian components (X and Y):

$$R_x = \sum F_x = F_1 \cos \theta_1 + F_2 \cos \theta_2 + \cdots + F_n \cos \theta_n \quad (5)$$

$$R_y = \sum F_y = F_1 \sin \theta_1 + F_2 \sin \theta_2 + \cdots + F_n \sin \theta_n \quad (6)$$

Magnitude of Resultant:

$$R = \sqrt{R_x^2 + R_y^2} \quad (7)$$

Direction (inclination with positive x-axis):

$$\theta = \tan^{-1} \left| \frac{R_y}{R_x} \right| \quad (\text{adjusted for correct quadrant using signs of } R_x, R_y) \quad (8)$$

1.6.2 B. Non-Concurrent Force System

In non-concurrent coplanar systems, forces do not intersect at a common point. The system reduces to:

1. A resultant force $\vec{R} = R_x\hat{i} + R_y\hat{j}$ acting at an arbitrarily chosen reference point O .
2. A resultant couple moment $M_O = \sum M_{O,i}$.

Simplest Resultant of Non-Concurrent Systems

There are three fundamental cases:

1. **Single Resultant Force** ($R \neq 0$): The force-couple system at O ($\vec{R}$ and M_O) can be further reduced to a **single force** $\vec{R}$ acting at a perpendicular distance d from O such that:

$$d = \frac{|M_O|}{R} \quad (9)$$

The line of action intercepts the coordinate axes at:

$$x\text{-intercept: } x_R = \frac{M_O}{R_y} \quad (\text{setting } y = 0), \quad y\text{-intercept: } y_R = -\frac{M_O}{R_x} \quad (\text{setting } x = 0) \quad (10)$$

2. **Pure Couple** ($R = 0$ but $M_O \neq 0$): The system cannot be reduced to a single force. The simplest resultant is a pure couple $C = M_O$, which is free to act anywhere in the plane.
3. **Complete Equilibrium** ($R = 0$ and $M_O = 0$): The system is completely balanced with zero net translational and rotational tendency.

1.7 1.7 Distributed Force Systems

Loads applied over an area, line, or volume are called distributed loads. In 2D beam analysis, line loads $w(x)$ (expressed in N/m or kN/m) are most common.

- **Total Resultant Force** (W): Equal to the area under the load diagram:

$$W = \int_0^L w(x) dx \quad (11)$$

- **Point of Action** ($\bar{x}$): Passes through the centroid of the loading diagram:

$$\bar{x} = \frac{\int_0^L x \cdot w(x) dx}{\int_0^L w(x) dx} \quad (12)$$

Load Distribution Type	Load Profile $w(x)$	Resultant Force (W)	Centroid Location ($\bar{x}$)
Uniformly Distributed (UDL)	$w(x) = w_0 = \text{const}$	$W = w_0 \cdot L$	$\bar{x} = L/2$
Uniformly Varying (UVL - Triangle)	$w(x) = \frac{w_0}{L}x$ (zero at $x = 0$)	$W = \frac{1}{2}w_0L$	$\bar{x} = \frac{2}{3}L$ (i.e., $\frac{L}{3}$ from base)
Trapezoidal Load	w_1 to w_2	$W = \left(\frac{w_1 + w_2}{2}\right)L$	Split into rectangle (w_1) and triangle

Table 1: Common 2D Distributed Force Configurations

1.8 1.8 Free Body Diagrams (FBD)

Definition and Importance

A **Free Body Diagram (FBD)** is a completely isolated sketch of the mechanical body or system under consideration, stripped of all surrounding physical supports, connections, and constraints, showing **all** external forces and moments applied by the removed environment.

Standard Support Reactions in 2D:

- **Frictionless Roller / Rocker Support:** Constrains motion perpendicular to contact surface $\Rightarrow$ 1 unknown reaction force (R_y normal to surface).
- **Hinged / Pinned Support:** Prevents translation in both X and Y directions, allows free rotation $\Rightarrow$ 2 unknown independent reaction forces (R_x, R_y).
- **Fixed / Built-in Support (Cantilever):** Prevents all translation and rotation $\Rightarrow$ 3 unknown reactions (R_x, R_y , and reaction moment M_R).
- **Flexible Cable / Rope / Link:** Force is strictly tensile and acts along the direction of the centerline of the cable $\Rightarrow$ 1 unknown tension T .
- **Smooth Surface Contact:** Compressive normal reaction force N perpendicular to the tangent of contact.

1.9 1.9 Equilibrium and Equations of Equilibrium

A rigid body is in static equilibrium when it has zero linear acceleration and zero angular acceleration.

General Equations of Equilibrium in 2D Coplanar System

For any general non-concurrent, non-parallel 2D force system:

$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum M_O = 0 \quad (\text{about any arbitrary point } O) \quad (13)$$

Alternative Sets of Independent Equilibrium Conditions:

1. $\sum F_x = 0, \quad \sum M_A = 0, \quad \sum M_B = 0$ (provided line AB is not perpendicular to the x -axis).
2. $\sum M_A = 0, \quad \sum M_B = 0, \quad \sum M_C = 0$ (provided points A, B, C are non-collinear).

Special Equilibrium Theorems for Concurrent Forces:

- **Two-Force Principle:** If a body is held in equilibrium by only two forces, the forces must be equal in magnitude, opposite in sense, and collinear.
- **Three-Force Principle:** If a body is in equilibrium under three coplanar forces, their lines of action must either be **concurrent** or **mutually parallel**.
- **Lami's Theorem:** If three coplanar concurrent forces act on a particle in equilibrium, each force is proportional to the sine of the angle between the other two forces:

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma} \quad (14)$$

where α, β, γ are the angles opposite to forces F_1, F_2, F_3 respectively.

1.10 1.10 Typical Applications of 2D Force Systems

- **Simply Supported and Overhanging Beams:** Determining reaction forces at supports under point loads, distributed loads, and applied concentrated moments.
- **Sphere Packing / Cylinders in Troughs:** Applying Lami's theorem and equilibrium equations to find contact forces between rolling cylinders and inclined walls.
- **Pin-Jointed Trusses (Method of Joints and Method of Sections):** Determining internal member axial forces (tensile or compressive) ensuring joint equilibrium $\sum F_x = 0, \sum F_y = 0$.

- **Cranes, Ladders, and Mechanical Linkages:** Evaluating pin reactions, cable tensions, and critical overturning moments.

2 Part 2: Friction

2.1 2.1 Introduction to Friction

When two rough surfaces are in contact and an external tangential effort is applied tending to cause relative sliding, a resisting force develops tangential to the interface called **friction**.

- **Microscopic Origin:** Even finely polished surfaces possess microscopic peaks and valleys (asperities). When pressed together, the real contact area is a tiny fraction of the apparent area, causing microscopic cold-welding and mechanical interlocking of asperities.
- **Types of Friction:**
 - *Dry Friction (Coulomb Friction):* Occurs between unlubricated contacting surfaces of rigid bodies.
 - *Fluid Friction:* Resistance encountered between layers of viscous fluids moving at different velocities.
 - *Skin / Boundary Friction:* Resistance encountered when a thin lubricant layer fails to completely separate surfaces.
 - *Rolling Resistance:* Resistance when a round cylinder or sphere rolls over a deformable surface.

2.2 2.2 Friction Regimes and Limiting Friction

Consider a block of weight W resting on a horizontal surface subjected to an increasing applied horizontal pull P :

1. **Static Regime** ($P < F_s$): As P increases, the frictional force F increases identically to maintain equilibrium ($F = P$). The body remains at rest.
2. **Limiting Friction** ($F = F_{\max} = F_s$): The maximum frictional resistance that can be mobilized just before sliding impends. This is the condition of **impending motion**.
3. **Kinetic / Dynamic Regime** ($P > F_s$): Motion commences. The interlocking asperities shear off or ride over one another, causing the friction force to drop to a slightly lower, virtually constant value $F_k < F_s$.

2.3 2.3 Laws of Coulomb (Dry) Friction

Formulated empirically by Guillaume Amontons (1699) and Charles-Augustin de Coulomb (1781):

1. The limiting static friction force is directly proportional to the normal contact reaction N :

$$F_{\text{limiting}} = \mu_s N \quad (15)$$

and for kinetic friction during steady sliding:

$$F_k = \mu_k N \quad (\text{where } \mu_k < \mu_s) \quad (16)$$

2. The total friction force is independent of the apparent nominal area of contact.
3. The friction force always acts tangential to the contact surfaces and in a direction strictly opposing the relative or impending motion.
4. For moderate speeds, kinetic friction is independent of sliding velocity.

2.4 Angle of Friction, Cone of Friction, and Angle of Repose

2.4.1 A. Angle of Friction (ϕ)

The resultant reaction $\vec{R}$ exerted by the surface on the body is the vector sum of normal reaction $\vec{N}$ and friction force $\vec{F}$:

$$\vec{R} = \vec{N} + \vec{F}, \quad R = \sqrt{N^2 + F^2} \quad (17)$$

The angle ϕ that the total reaction $\vec{R}$ makes with the normal $\vec{N}$ is given by:

$$\tan \phi = \frac{F}{N} \implies \tan \phi_s = \frac{F_s}{N} = \mu_s \implies \phi_s = \tan^{-1}(\mu_s) \quad (18)$$

ϕ_s is known as the **angle of limiting static friction**.

2.4.2 B. Cone of Friction

If impending motion can occur in any arbitrary direction in the tangent contact plane, the resultant reaction vector $\vec{R}$ describes an inverted right circular cone of vertex half-angle ϕ_s around the normal $\vec{N}$.

- If the applied resultant falls **inside** the cone of friction, the body remains stationary (self-locking).
- If the applied resultant lies **on the surface** of the cone, motion is impending.
- Equilibrium is impossible if the line of action falls outside the cone.

2.4.3 C. Angle of Repose (θ or α)

The angle of repose is the maximum inclination angle of a rough plane with the horizontal at which a body placed upon it will remain at rest without sliding down under its own weight.

Derivation of Equivalence

Consider a block of mass m on an incline θ :

- Normal equilibrium: $N = mg \cos \theta$
- Tangential equilibrium at impending downward sliding: $F = mg \sin \theta = \mu_s N = \mu_s mg \cos \theta$
- Dividing: $\tan \theta = \mu_s = \tan \phi_s \implies \theta = \phi_s$

Conclusion: The angle of repose is numerically identical to the angle of limiting friction.

2.5 Equilibrium of Bodies Involving Dry Friction

When solving problems involving dry friction, classify the problem into one of three distinct categories:

1. **Equilibrium Assured (No impending motion):** Frictional force is an unknown self-adjusting variable. $F < \mu_s N$. Determine F strictly via equations of statics ($\sum F_x = 0, \sum F_y = 0, \sum M = 0$). Check whether $F \leq \mu_s N$.
2. **Motion is Impending at Known Contact Surface:** The maximum friction is mobilized. Use $F = \mu_s N$ as an auxiliary equation alongside equilibrium equations.
3. **Impending Motion Type Unknown (Sliding vs. Tipping/Overturning):**
 - *Case A (Sliding):* Set $F = \mu_s N$ and compute P_{slide} .
 - *Case B (Tipping):* Tipping occurs about the pivot edge when the normal reaction shifts to the extreme corner ($x = b/2$). Set $\sum M_{\text{corner}} = 0$ to find P_{tip} .
 - *Governing mode:* The smaller load $P = \min(P_{\text{slide}}, P_{\text{tip}})$ dictates which failure mode occurs first!

2.6 2.6 Belt Friction (Flexible Flat Belts and V-Belts)

When a flexible inextensible belt or rope passes over a cylindrical rough drum or pulley, friction between the belt and drum surface allows torque and power transmission.

Derivation of Flat Belt Friction Formula (Capstan Equation)

Consider a small differential element of belt subtending angle $d\theta$:

- Tension increases from T to $T + dT$ along the arc.
- Normal reaction on element: dN . Frictional force: $dF = \mu dN$.
- Resolving along the normal direction:

$$dN = (T + dT) \sin\left(\frac{d\theta}{2}\right) + T \sin\left(\frac{d\theta}{2}\right) \approx T d\theta \quad (\text{since } \sin(d\theta/2) \approx d\theta/2) \quad (19)$$

- Resolving along the tangential direction (impending slip towards tighter side T_1):

$$(T + dT) \cos\left(\frac{d\theta}{2}\right) - T \cos\left(\frac{d\theta}{2}\right) - dF = 0 \implies dT = dF = \mu dN = \mu T d\theta \quad (20)$$

- Separating variables and integrating across angle of lap β (in radians):

$$\int_{T_2}^{T_1} \frac{dT}{T} = \mu \int_0^\beta d\theta \implies \ln\left(\frac{T_1}{T_2}\right) = \mu\beta \quad (21)$$

$$\frac{T_1}{T_2} = e^{\mu\beta} \quad (22)$$

Definitions:

- T_1 = Tension on the tight side (always larger, in direction opposing motion/impending slip).
- T_2 = Tension on the slack side.
- μ = Coefficient of friction between belt and drum.
- β = Total angle of contact or wrap (**must be in radians!**).

2.6.1 Extension to V-Belts and Ropes in Grooved Pulleys

For a V-belt running in a grooved pulley with groove angle 2α , the normal forces act on the inclined sides of the groove. Resolving gives the effective coefficient of friction:

$$\mu' = \frac{\mu}{\sin \alpha} \implies \frac{T_1}{T_2} = e^{\frac{\mu\beta}{\sin \alpha}} \quad (23)$$

Because $\sin \alpha < 1$, $\mu' > \mu$, which provides significantly higher frictional grip (wedging effect) without slipping.

2.7 2.7 Typical Applications of Friction

1. **Ladder Friction:** A ladder leaning against a rough/smooth wall. Requires applying $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$, and setting $F \leq \mu N$ at both foot and wall contact points.
2. **Wedge Friction:** Wedges are used to lift heavy loads or split objects. Each contact interface develops normal force N_i and friction force $F_i = \mu N_i$. Solved via simultaneous equilibrium equations or force polygons.
3. **Square-Threaded Screws / Screw Jacks:** Modeled as an inclined plane wrapped around a cylinder of helix angle $\alpha = \tan^{-1}\left(\frac{p}{\pi d_m}\right)$. Effort to raise load: $P = W \tan(\alpha + \phi_s)$. Condition for self-locking: $\phi_s > \alpha$.

4. **Band Brakes and Capstans:** Utilizing the belt friction equation to calculate the braking torque $T_B = (T_1 - T_2)R$ or the holding force of mooring ropes around bollards.

3 Part 3: Representative Analytical Solved Frameworks

3.1 Problem 1: Non-Concurrent Force System Reduction

Problem Statement: Four forces act along the sides of a $2\text{ m} \times 2\text{ m}$ square $ABCD$ as follows: 10 N along AB (A to B), 20 N along BC (B to C), 30 N along CD (C to D), and 40 N along DA (D to A). Reduce the system to a single resultant force and find its line of action relative to origin $A(0,0)$.

Step-by-step Solution:

1. **Coordinates of Vertices:** $A(0,0)$, $B(2,0)$, $C(2,2)$, $D(0,2)$.
2. **Resolution of Forces:**

$$\begin{aligned}\vec{F}_{AB} &= 10\hat{i} \quad (\text{along } y = 0) \\ \vec{F}_{BC} &= 20\hat{j} \quad (\text{along } x = 2) \\ \vec{F}_{CD} &= -30\hat{i} \quad (\text{along } y = 2) \\ \vec{F}_{DA} &= -40\hat{j} \quad (\text{along } x = 0)\end{aligned}$$

3. **Resultant Components:**

$$\begin{aligned}R_x &= \sum F_x = 10 - 30 = -20\text{ N} \\ R_y &= \sum F_y = 20 - 40 = -20\text{ N} \\ R &= \sqrt{(-20)^2 + (-20)^2} = \sqrt{800} = 20\sqrt{2} \approx 28.28\text{ N} \\ \theta &= \tan^{-1}\left(\frac{-20}{-20}\right) = 225^\circ \quad (\text{pointing into 3rd quadrant})\end{aligned}$$

4. **Moment about Origin $A(0,0)$ (Counter-Clockwise positive):**

- Force AB passes through $A \implies M_1 = 0$
- Force BC : $+20 \times 2 = +40\text{ N}\cdot\text{m}$ (CCW)
- Force CD : $+30 \times 2 = +60\text{ N}\cdot\text{m}$ (CCW)
- Force DA passes through $A \implies M_4 = 0$
- Total $M_A = 40 + 60 = +100\text{ N}\cdot\text{m}$ (CCW)

5. **Line of Action of Resultant:** Using Varignon's Theorem: $M_A = xR_y - yR_x$

$$100 = x(-20) - y(-20) \implies -20x + 20y = 100 \implies \mathbf{y - x = 5}$$

x -intercept (set $y = 0$): $x = -5\text{ m}$. y -intercept (set $x = 0$): $y = 5\text{ m}$.

3.2 Problem 2: Classical Ladder Friction Equilibrium

Problem Statement: A uniform ladder of mass $M = 30\text{ kg}$ and length $L = 5\text{ m}$ rests against a smooth vertical wall at B and on a rough horizontal floor at A . The coefficient of static friction at floor is $\mu_s = 0.3$. Determine the minimum inclination angle $\theta_{\min}$ of the ladder with the floor before it starts slipping.

Step-by-step Solution:

1. **Identify Acting Forces:**

- Weight $W = Mg = 30 \times 9.81 = 294.3\text{ N}$ acting downwards at midpoint G ($L/2$ from A).

- Smooth wall at $B \implies$ strictly horizontal normal reaction N_B perpendicular to wall.
- Rough floor at $A \implies$ vertical normal reaction N_A and friction force F_A pointing towards wall (to resist outward slip).

2. Equilibrium Equations:

$$\begin{aligned}\sum F_y = 0 &\implies N_A - W = 0 \implies N_A = W = 294.3 \text{ N} \\ \sum F_x = 0 &\implies F_A - N_B = 0 \implies N_B = F_A\end{aligned}$$

3. **Limiting Condition:** At impending slipping: $F_A = \mu_s N_A = 0.3 \times 294.3 = 88.29 \text{ N} \implies N_B = 88.29 \text{ N}$.

4. **Moment Equilibrium about Foot A ($\sum M_A = 0$):**

$$\begin{aligned}\sum M_A = 0 &\implies N_B(L \sin \theta) - W \left(\frac{L}{2} \cos \theta \right) = 0 \\ N_B \sin \theta &= \frac{W}{2} \cos \theta \\ \tan \theta &= \frac{W}{2N_B} = \frac{W}{2(\mu_s W)} = \frac{1}{2\mu_s} \\ \tan \theta &= \frac{1}{2 \times 0.3} = \frac{1}{0.6} = 1.6667 \\ \theta_{\min} &= \tan^{-1}(1.6667) \approx 59.04^\circ\end{aligned}$$

3.3 Problem 3: Capstan / Belt Friction Application

Problem Statement: A sailor wraps a rope around a stationary cylindrical capstan 2.5 full turns ($\beta = 2.5 \times 2\pi = 5\pi \text{ rad}$). If the coefficient of friction is $\mu = 0.25$, find the pull force T_2 required to hold an ocean liner exerting a pull of $T_1 = 50 \text{ kN}$.

Step-by-step Solution:

1. Angle of wrap: $\beta = 2.5 \times 2\pi = 5\pi \approx 15.708 \text{ rad}$.
2. Using belt friction equation:

$$\frac{T_1}{T_2} = e^{\mu\beta} = e^{0.25 \times 15.708} = e^{3.927} \approx 50.75$$

3. Calculating slack side tension:

$$T_2 = \frac{T_1}{50.75} = \frac{50\,000 \text{ N}}{50.75} \approx 985.2 \text{ N} \quad (\approx 0.985 \text{ kN})$$

4. **Physical Insight:** A modest human force of less than 100 kgf holds a massive 50 kN ship entirely due to the exponential grip of capstan friction!

4 Part 4: HBTU Semester Exam Quick Reference & FAQs

High-Yield HBTU Examination Topics & Common Pitfalls

- **Lami's Theorem Prerequisites:** Ensure the forces are concurrent, coplanar, and strictly three in number. Check direction arrows: all three vectors must point away from (or towards) the concurrent point.
- **Varignon's Theorem Signs:** Always declare a standard sign convention (e.g., Counter-Clockwise as positive) before writing moment equations.
- **Angle of Wrap in Belt Friction:** Always convert rotations/degrees into **radians** ($\beta =$

$\theta^\circ \times \frac{\pi}{180}$ or $\beta = 2\pi n$) before substituting into $e^{\mu\beta}$.

- **Limiting Friction Assumption:** Do NOT blindly set $F = \mu N$. State clearly whether the body is in static equilibrium ($F \leq \mu_s N$) or impending motion ($F = \mu_s N$).
- **FBD Completeness:** An exam answer without a clear Free Body Diagram typically loses 30–40% of marks in HBTU evaluations. Always explicitly sketch reaction components.